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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431, BSc Scheme in COMP & AI-61435

Subject : COMP3438 System Programming

Group : 1011, 1012, 141

Session : 2023 / 2024 Semester I

Date : 8 Dec 2023 Time : 12:30-14:30

Time Allowed: 2 Hours

Subject Lecturer: Dr WANG Qixin / Dr SAXENA Divya

This question paper has ___8___ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

1. Closed books and notes.
2. Answer ALL questions.

Do not turn this page until you are told to do so!

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Section A

Question 1 [15%]

Read the following source code led1.c for implementing an LED driver, and led1app.c for the corresponding user level application

led1.c

```
00 #include <linux/module.h>
01 #include <linux/cdev.h>
02 #include <linux/kdev_t.h>
03 #include <linux/init.h>
04 #include <linux/fs.h>
05 #include <linux/types.h>
06 #include <asm/uaccess.h>
07 #include <mach/gpio.h>
08 #include <plat/gpio-cfg.h>
09
10 #define DRIVER_NAME "LED1 Driver"
11 #define LED1 S5PV210_GPJ2(0)
12 #define LED_ON 0
13 #define LED_OFF 1
14 #define N_D 1
15 #define S_N 1
16 static dev_t devno;
17 static int major;
18 static struct cdev form;
19
20 static int led1_open(struct inode *inode,
   struct file *fp) {
21     printk("Device " DRIVER_NAME " open.\n");
22     return 0;
23 }
24
25 static ssize_t led1_write(struct file *fp,
26 const char _____ *buf, size_t count, loff_t *offset) {
27     char led_status;
28     int ret;
29     ret = _____(&led_status, _____,
           sizeof(led_status));
30     if (ret) {
31         printk(
               "Fail to copy data from the user space "
               "to the kernel space.\n");
32         return 0;
33     }
34     printk("Write: led = 0x%x.\n", led_status);
```

(to be continued on the next page ...)

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(continued from the previous page.)

```
35      if (led_status > '0')
36          gpio_set_value(LED1, LED_ON);
37      else
38          gpio_set_value(LED1, LED_OFF);
39      return sizeof(led_status);
40  }
41
42  static int led1_release(struct inode *inode,
    struct file *fp) {
43      printk("Device " DRIVER_NAME " release.\n");
44      return 0;
45  }
46
47  static struct file_operations fops = {
48      owner: ______,
49      open: ______,
50      write: ______,
51      release: ______,
52  };
53
```

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(continued from the previous page)

```
54 static int __init led1_init(void) {
55     int ret;
56     s3c_gpio_cfgpin(LED1, S3C_GPIO_OUTPUT);
57     gpio_set_value(LED1, LED_OFF);
58
59     ret = alloc_chrdev_region(&devno, S_N, N_D,
           DRIVER_NAME);
60     if (ret < 0) {
61         printk("Device " DRIVER_NAME
               " cannot get major number.\n");
62         return ret;
63     }
64     major = _____(devno);
65     printk("Device " DRIVER_NAME
           " initialized (Major number = %d).\n", major);
66
67     cdev_init(&form, &fops);
68     form.owner = THIS_MODULE;
69     ret = _____(&form, devno, N_D);
70     if (ret) {
71         printk("Device " DRIVER_NAME
               " register fail.\n");
72         return ret;
73     }
74     return 0;
75 }
76
77 static void __exit led1_exit(void) {
78     _____(&form);
79     unregister_chrdev_region(devno, N_D);
80     printk("Device " DRIVER_NAME " unloaded.\n");
81 }
82
83 module_init(_____);
84 module_exit(_____);
85
86 MODULE_LICENSE("GPL");
87 MODULE_AUTHOR("Qixin Wang");
88 MODULE_DESCRIPTION("LED1 Driver Demo");
89
```

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led1app.c

```
00 #include <stdio.h>
01 #include <sys/types.h>
02 #include <sys/stat.h>
03 #include <fcntl.h>
04 #include <unistd.h>
05
06 int main() {
07     int fd;
08     char input;
09
10     fd = open("/dev/led1", _____);
11     if (fd == -1) {
12         perror("Fail to open /dev/led1.\n");
13         return -1;
14     }
15
16     while (1) {
17         printf("User: input \'0\' to turn off the LED, ");
18         printf("\'1\' to turn on the LED, ");
19         printf("\'2\' to quit.\n");
20         scanf(" %c", _____);
21         if (input == '2') break;
22         _____(fd, &input, 1);
23     }
24     close(fd);
25     return 0;
26 }
```

Suppose all source codes' identifiers and printed messages are truthful. Please fill in the blanks in the source codes respectively with one of the following choices (these choices are listed alphabetically for your convenience).

A. user
B. &buf
C. &input
D. MAJOR
E. MODULE_AUTHOR
F. MODULE_DESCRIPTION
G. MODULE_LICENSE
H. O_CREAT
I. O_RDONLY
J. O_WRONLY
K. THIS_MODULE

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L. buf
M. cdev_add
N. cdev_del
O. copy_from_user
P. copy_to_user
Q. input
R. led1_exit
S. led1_init
T. led1_open
U. led1_release
V. led1_write
W. read
X. write

(a) For the blank of led1.c line 26, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(b) For the first blank of led1.c line 29, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(c) For the second blank of led1.c line 29, we should fill in ___ (choose one from 'A', 'B', ..., 'X').
(d) For the blank of led1.c line 48, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(e) For the blank of led1.c line 49, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(f) For the blank of led1.c line 50, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(g) For the blank of led1.c line 51, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(h) For the blank of led1.c line 64, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(i) For the blank of led1.c line 69, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(j) For the blank of led1.c line 78, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(k) For the blank of led1.c line 83, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(l) For the blank of led1.c line 84, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(m) For the blank of led1app.c line 10, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(n) For the blank of led1app.c line 20, we should fill in _____ (choose one from 'A', 'B', ..., 'X').
(o) For the blank of led1app.c line 22, we should fill in _____ (choose one from 'A', 'B', ..., 'X')

Question 2 [3%]

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Question 3 [3%]

Question 4 [3%]

Question 5 [6%]

Write a Regular Expression for a language consisting of strings over {a, b} that contain an even number of a's.

Section B

Question 6 [10%]

Consider a fragment of a high-level programming language:

```
int x = 10;
int y = x * 2;
int z = y + 5;
```

(a) Translate the given code fragment into intermediate code (such as Three-Address Code). [4%]

(b) Discuss how a symbol table is used during this translation process. [4%]

(c) Apply basic code optimization techniques to the intermediate code generated in Part (a). [2%]

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Question 7 [10%]

Consider the following grammar for a section of a language that handles variable assignments and arithmetic operations:

1. S → V = E | V = N
2. E → V + N | V - N
3. V → id | id [N]
4. N → num

In this grammar:
Non-Terminals: V (Variable), E (Expression), N (Number)
Terminals: =, +, -, id, [,], num
Perform left factoring on this grammar.

Question 8 [20%]

Consider the following Regular Expression (RE):
RE = ab* + a

(a) Construct an NFA directly from the given RE without using ε-transitions. [10%]
(b) Convert the directly constructed NFA into an equivalent DFA. [10%]

Section C

Question 9 [30%]
Consider the following grammar:

1. S → Aa | bAc | Bd
2. A → d | ε
3. B → dc | ε

In this grammar:
ε represents NULL;
Non-Terminals: S, A, B;
Terminals: a, b, c, d, ε;

(a) Calculate the First sets for each non-terminal in the grammar. [10%]
(b) Calculate the Follow sets for each non-terminal in the grammar. [10%]
(c) Construct the LL(1) parsing table for the grammar and determine if the grammar is LL(1). [10%]

** END **